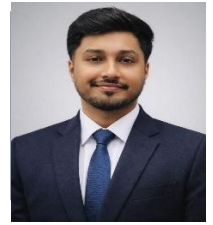


MD TANVIR ARABI

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LinkedIn: <https://www.linkedin.com/in/tanvirarabi/> | **GitHub:** <https://github.com/Tanvirarabi>



Career Summary

As a Computer Science Engineering student with a strong foundation in machine learning, data analysis, and software development. I am proficient in Python, C++, HTML, CSS, and JavaScript. Passionate about advanced machine learning, deep learning, and image processing techniques, with a focus on real-world applications.

Experience

Software Engineering – AI/ML

1st Feb 2026-Present

Project 1: Smart Dustbin Garbage Monitoring System- (Bangladesh Railway)

- **Python (Django)** : Developed robust backend APIs using Django for seamless data management and integration with deep learning models.
- **Celery**: Implemented asynchronous task management with Celery to handle long-running processes such as data processing and model inference in a scalable way.
- **MQTT**: Utilized MQTT protocol for real-time communication and IoT device integration, ensuring fast and reliable message delivery between clients and the server.

Project 2: Develop & Design Five ERP Modules: HR, PM, Procurement, Inventory, Accounting.(Full-Stack)

- **Core System Development**: Built 5 full-featured ERP modules (HR, Project Management, Procurement, Inventory, and Accounting) using Django and React.js.
- **API & Database Management**: Designed clean REST APIs and managed MySQL databases to ensure fast, reliable data processing across all modules.
- **Frontend & Backend Integration**: Connected interactive React user interfaces with Python backend services for seamless user workflows.
- **End-to-End Delivery**: Managed the complete software development lifecycle from initial database design to final feature implementation.

Project 3: Smart Livestock Management System – (Python, Django, Deep Learning)

- Developed a web-based system to identify individual cows using muzzle patterns.
- Built a deep learning model to estimate actual cow weight from images.
- Enabled practical farm management through automated identification and weight monitoring.

Datasoft Manufacturing & Assembly Limited

Link: <https://dma.com.bd/>

Education

American International University-Bangladesh (AIUB)

Bachelor of Science in Computer Science and Engineering (B.Sc. in CSE)

2022 – 2026

Major in Information System -

(Machine Learning, Artificial Intelligence, Deep Learning, Data Science, Natural Language Processing, Network Security, Software Engineering, Webtechnologies)

CGPA- 3.24

Cantonment Public School & College, Rangpur
Higher Secondary School Certificate (HSC), Science
GPA – 5.00

2018- 2020

Srijony High School, Joypurhat
Secondary School Certificate (SSC), Science
GPA – 5.00

2016– 2017

Publications

- **Forecasting Potential Dengue Infection from Clinical and Environmental Data Using a Machine Learning based Meta model** **Dec-2025**
5th International Conference on Electrical, Computer & Telecommunication Engineering (ICECTE, RUET 2026)
[Link: https://icecte.ruet.ac.bd/](https://icecte.ruet.ac.bd/)

Manual & Automation Testing with Selenium, Playwright, Postman- *Ostad Platform*

- **Automated UI & functional and Nonfunctional Testing (API, Locator, Database), Bugs reporting**
1. <https://www.saucedemo.com/>
2. <https://demo.evershop.io/>
E-commerce site using **Selenium and Node.js**, ensuring robust cross-browser validation.
GitHub: Repository (Test-With-Selenium-<https://github.com/Tanvirarabi>)

University Projects

- **Forecasting Potential Dengue Infection from Clinical and Environmental Data Using a Machine Learning based Meta model (Machine Learning)**
Description: Build a meta-model using four base models: ANN, KNN, XGBoost, and Decision Tree, to predict the risk of dengue based on a Kaggle dataset containing 1000 clinical and environmental data points.
Tools/Frameworks: Python Language, NumPy, Pandas, PyTorch, Seaborn, scikit-learn, matplotlib
GitHub-Link: <https://github.com/Tanvirarabi>
- **Real-vs-Fake-News-Detection-NLP-Naive-Bayes (Natural Language Processing)**
Description: Preprocess a real vs fake news text dataset with approximately 9000 entries (51% fake and 49% real) using tokenization technique and then apply the Naive Bayes algorithm for Natural Language Processing.
Tools/Frameworks: Python Language, Pandas, NumPy, seaborn, scikit-learn, matplotlib
GitHub-Link: <https://github.com/Tanvirarabi>
- **Data-Science-Approaches-for-Imbalanced-Dataset-Preprocessing (Data Science)**
Description: Preprocess the imbalanced dataset by handling missing values and preparing the dataset to be clean and trainable for model training using R language.
Tools: R language.
GitHub-Link: <https://github.com/Tanvirarabi>
- **Air-Quality-Index-Web-technologies (Full-Stack)**
Description: Develop a web form based on the Air Quality Index (AQI) using HTML, CSS, and JavaScript for the frontend, and PHP and MySQL for the backend, creating a full-stack project
Tools: HTML, CSS, JavaScript, PHP
GitHub-Link: <https://github.com/Tanvirarabi>
- **Advance-Database-Management-System-Project(PL/SQL)**

Description: Create a system for updating multiple employee records to track Oracle 10g and prepare the CRUD operations

Tools: Oracle 10g.

GitHub-Link: <https://github.com/Tanvirarabi>

- **Hotel Management System(C# , .net)**

Description: Develop a web form based Hotel Management System to Integrate monitoring, booking, payment, Admin & User Pannel Also Hotel stuffs information & CURD operation .

Language: C#, MySQL

GitHub-Link: <https://github.com/Tanvirarabi>

Technical Skills

Languages: Python, HTML, CSS, JavaScript, PHP, C++, R, MySQL, PL/SQL, WordPress.

Tools/Frameworks: MySQL, Oracle 10g, Postman, Selenium, Playwright, Trello, GitLab, TensorFlow, NumPy, Pandas, seaborn, keras, Scikit-learn, GitHub

Domains: Machine Learning, Deep Learning, Data Science, Natural Language Processing, SQA, Web developer.

Soft Skills

- Leadership
- Business Development
- Risk Management
- Analytical Problem-Solving
- Effective Communication
- Team Collaboration
- Adaptability and Quick Learning
- Presenter(5th, ICECTE-RUET)

Reference

- **Tonny Shekha Kar**
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American International University-Bangladesh
Email: tonny.kar@aiub.edu
- **Mubasshar-Ul-Ishraq Tamim**
Lecturer, FST
American International University-Bangladesh
Email: mubasshar@aiub.edu